Extensions of the split affine residues

Human Compression project

Abstract

We determine every rank-four residually weakly primitive extension of the specified residues R7 and R9 in the fixed higher-rank Suzuki paper. The proof uses the actual prescribed intersections and does not assume that paper's classification of all rank-three residues. The main step is a subfield containment lemma for an affine subgroup whose scalar complement generates its coefficient field.

1 Conventions and structural premises

Write $S_f = \mathrm{Sz}(2^f)$ for the standard subfield group in S_N, where $f \mid N$ and both exponents are odd. All simple subfield groups below have $f \geq 3$. Write H_m for the subgroup of order $m \mid 2^N - 1$ in the standard split torus. Its Weyl involution w inverts it. Write Z_{2^f} for the involution centre of the standard Sylow two-subgroup Q_{2^f}. In particular, Z_{2^f} has the additive coordinates of $\mathbb{F}_{2^f}$. Put $D_{2m} = H_m\langle w\rangle$.

We use the classical Suzuki matrix model, its subgroup classification, and conjugacy of subfield groups and split tori, as in the accepted preliminary papers. Specifically, a proper nonsoluble subgroup is a simple subfield group. The other maximal subgroup types are a point stabilizer, a split torus normalizer, and one of the two nonsplit torus normalizers. A nonidentity odd-order element lies in a unique torus; $N_{S_N}(H_m)$ is the full split normalizer for $m > 1$. Simple subfield groups are self-normalizing. A point stabilizer has shape $Q_q : H_{q-1}$, where $|Z_q| = |Q_q/Z_q| = q$ and $[Q_q, Q_q] = Z_q$. The torus acts on both additive quotients by scalar powers coprime to $q - 1$. A nonidentity involution fixes a unique ovoid point. These are explicit structural premises, not consequences of the classification being corrected.

The source's residual weak primitivity condition says that, for every proper subset J of the type set, some remaining intersection $G_{J\cup\{i\}}$ is maximal proper in G_J. It implies that every nested intersection is strict. Indeed, if $G_J \leq G_i$ for $i \notin J$, then $G_{I\setminus\{i\}} \leq G_J \leq G_i$, contradicting maximal proper inclusion at $I \setminus \{i\}$. We use this consequence explicitly; no geometric flag transitivity is assumed.

Lemma 1. *A fixed embedded simple subfield group S_e is contained in at most one subfield group of any prescribed degree f.*

Proof. Suppose a conjugate of S_f contains S_e. Transporting to S_f and using conjugacy of its subfield groups, adjust the transporter so that it normalizes S_e. Its ambient normalizer is S_e, which is already contained in the standard S_f. Thus the conjugate is S_f itself. □

2 A field-generating affine complement

Lemma 2. *Let $s = 2^f \geq 8$, let $f \mid N$, and let $m \mid s - 1$ such that H_m generates $\mathbb{F}_s$ as a field over $\mathbb{F}_2$. Any simple subfield group containing either D_{2m} or $Z_s : H_m$ contains the standard S_f.*

Proof. Let T have degree v. First H_m is split in T. Indeed, the order of 2 modulo m is f, an odd number. If H_m were in a nonsplit torus, then $m \mid 2^{2v}+1$, so this order would divide $4v$ but not $2v$, which is impossible for an odd order. Consequently $m \mid 2^v-1$, and field generation gives $f \mid v$.

All subgroups of order m in split tori of S_v are conjugate. If $T = gS_vg^{-1}$ contains H_m, adjust g by an element of S_v so that it normalizes H_m. The ambient normalizer is the split dihedral group. Its reflection normalizes S_v, so we may take $g = h \in H_{q-1}$, where $q = 2^N$.

If $w \in T$, then $h^{-1}wh = h^{-2}w \in S_v$, whence $h^2 \in H_{2^v-1}$. All torus orders are odd, so $h \in H_{2^v-1}$ and $T = S_v$. This treats D_{2m}.

For the affine case, $T \cap Z_q = hZ_{2^v}h^{-1}$ has additive coordinate set $\lambda\mathbb{F}_{2^v}$, where λ is the scalar by which h acts on Z_q. Containment of Z_s implies $1 \in \mathbb{F}_s \subseteq \lambda\mathbb{F}_{2^v}$, hence $\lambda \in \mathbb{F}_{2^v}$. The central scalar exponent is coprime to $q-1$. Taking its inverse power shows that $h \in H_{2^v-1}$. Again $T = S_v$, which contains S_f. □

We also use the following acquired arithmetic fact. If $t = 2^e \geq 8$, a is an odd prime, and $n \mid (t^a-1)/(t-1)$ is prime, then $H_{(t-1)n}$ generates $\mathbb{F}_{t^a}$. If $n \nmid t-1$, the order of t modulo n is a. If $n \mid t-1$, then $n = a$ and the valuation of n in t^a-1 is exactly one larger than in $t-1$. In either case no proper intermediate field contains this cyclic subgroup. The full valuation proof, including overlap with $t-1$, is in the accepted paper *Scalar fields and split torus families.*

3 All extensions of the prescribed residue

Theorem 3. *Let a rank-four residually weakly primitive coset pregeometry in $G = \mathrm{Sz}(q)$ have $G_0 = \mathrm{Sz}(s)$ maximal in G. Suppose its prescribed G_0-residue is R7 or R9. In explicit coordinates put*

$$t = 2^e \geq 8, \quad s = t^a, \quad b = t-1, \quad m_0 = bn, \qquad n \mid (s-1)/b \text{ prime},$$

where a is prime. The prescribed lattice is

$$G_{01} = \mathrm{Sz}(t), \quad G_{02} = D_{2m_0}, \quad G_{03} = Z_s : H_{m_0},$$

$$G_{012} = D_{2b}, \quad G_{013} = Z_t : H_b, \quad G_{023} = H_{m_0}, \quad G_{0123} = H_b.$$

After one simultaneous conjugation, every extension has

$$q = t^{ac}, \quad G_1 = \mathrm{Sz}(v), \quad v = t^c, \qquad a \neq c \text{ primes},$$

and, for a prime $n' \neq n$ dividing $(v-1)/b$,

$$G_2 = D_{2bnn'}, \qquad G_3 = Z_q : H_{bnn'}.$$

All remaining intersections are

$$G_{12} = D_{2bn'}, \quad G_{13} = Z_v : H_{bn'}, \quad G_{23} = H_{bnn'}, \quad G_{123} = H_{bn'}.$$

Proof. We first justify the coordinates in the statement. Conjugate G_0 and its simple subfield G_{01} to the standard copies. The full split normalizer G_{012} in G_{01} may then be made $H_b\langle w\rangle$. The odd rotation subgroup of G_{02} lies in the unique ambient split torus containing H_b, so $G_{02} = H_{m_0}\langle w\rangle$. The group G_{03} is an affine subgroup of G_0 with elementary abelian radical of order s. Its radical is the centre of a Sylow two-subgroup of G_0. Since $G_{023} = H_{m_0}$, its fixed point is one of the two torus axis points. Interchanging them by w if necessary gives

$G_{03} = Z_s : H_{m_0}$. This also gives the displayed coordinates for G_{013} and all deeper prescribed intersections.

Since G_0 is maximal, $q = s^c$ for a prime c. The proper group G_1 contains the simple $\mathrm{Sz}(t)$, hence is itself a simple subfield group. Lemma 1 makes it standard. Its degree is a proper divisor of eac, contains e, and has gcd e with ea because $G_0 \cap G_1 = \mathrm{Sz}(t)$. Therefore its degree is ec, with $a \neq c$. Thus $G_1 = \mathrm{Sz}(v)$, $v = t^c$.

The scalar group H_{m_0} generates $\mathbb{F}_s$. Lemma 2 and maximality of S_{ea} show that no proper subfield overgroup of G_{02} or G_{03} can lie outside G_0. Strictness excludes containment of G_2 or G_3 in G_0. A point stabilizer cannot contain D_{2m_0}: the odd torus fixes its two axis points and the reflection interchanges them. Nonsplit torus normalizers are also excluded by their odd orders. Hence

$$G_2 = D_{2M}, \quad M \mid q-1, \qquad m_0 = \gcd(M, s-1), \quad m_1 := \gcd(M, v-1) > b.$$

Here $G_{12} = D_{2m_1}$ and $\gcd(m_0, m_1) = b$. The elementary abelian subgroup Z_s has order at least eight, excluding both sorts of torus normalizer as overgroups of G_3. Its fixed point is unique. Thus G_3 lies in the standard point stabilizer $B_q = Q_q : H_{q-1}$.

Write $G_{23} = H_h$. This is cyclic, since the reflection coset of G_2 is outside B_q. Strictness and the prescribed intersection with G_0 give $h > m_0$ and $\gcd(h, s-1) = m_0$. The field generated by H_h contains $\mathbb{F}_s$ and is not $\mathbb{F}_s$; since $[\mathbb{F}_q : \mathbb{F}_s] = c$ is prime, it is $\mathbb{F}_q$.

Put $N = G_3 \cap Q_q$. Its intersection with Z_q contains Z_s. The scalar action of H_h is irreducible on Z_q, so $N \cap Z_q = Z_q$. Its image on Q_q/Z_q is also invariant under this irreducible scalar action. Thus $N = Z_q$ or $N = Q_q$. The second case would make $G_0 \cap G_3$ contain Q_s, contradicting the prescribed central affine group. Therefore $N = Z_q$.

The quotient G_3/Z_q maps injectively into the cyclic complement of B_q/Z_q, and is therefore abelian. It contains the nontrivial scalar subgroup H_h. In the affine quotient $Q_q/Z_q : H_{q-1}$, a translation commuting with a nonidentity scalar must be zero. Every element of G_3/Z_q therefore has zero translation coordinate. Consequently

$$G_3 = Z_q : H_d, \qquad h = \gcd(M, d), \qquad m_0 \mid h \mid d.$$

Put $d_1 = \gcd(d, v-1)$. Both $m_0 < d$ and $d_1 < d$: the former follows from $h > m_0$, and the latter from field generation by H_{m_0}, since $\mathbb{F}_s \not\subseteq \mathbb{F}_v$. The intersections G_{03} and G_{13} have proper intermediate overgroups $Z_q : H_{m_0}$ and $Z_q : H_{d_1}$ in G_3. If $h < d$, then $G_{23} = H_h < H_d < G_3$ as well. This would violate residual weak primitivity at G_3. Hence $h = d$, so $d \mid M$.

We claim $d = M$. Suppose $d < M$. Then H_d is not maximal in D_{2M}, because $H_d < H_M < D_{2M}$. Residual weak primitivity at G_2 makes either D_{2m_0} or D_{2m_1} maximal. For odd dihedral groups this means M/m_0 or M/m_1 is prime: all overgroups of a fixed reflection are obtained by enlarging its cyclic rotation subgroup. If M/m_0 is prime, the strict divisor chain $m_0 < d \mid M$ gives $d = M$, a contradiction. Otherwise M/m_1 is prime. Since $m_0/\gcd(m_0, m_1) = n$ divides it, we have $M = nm_1$. As $m_0 \mid d < M = \mathrm{lcm}(m_0, m_1)$, one has $\gcd(d, m_1) < m_1$. In the residue at $G_{12} = D_{2m_1}$, the intersection with G_3 is then a proper subgroup of H_{m_1} and cannot be maximal. The intersection with G_0, namely D_{2b}, must be maximal. Thus m_1/b is prime. But $M/m_0 = m_1/b$, again forcing $d = M$ from $m_0 < d \mid M$. This contradiction proves the claim.

Now $G_{123} = H_{m_1}$. The final residual condition makes H_b maximal in it, so $m_1 = bn'$ for a prime n'. The equality $\gcd(m_0, m_1) = b$ gives $n' \neq n$. At $G_{23} = H_M$, at least one of H_{m_0} and H_{m_1} is maximal. The prime index must respectively be n' or n, because the other subgroup is contained in H_M. Therefore $M = bnn'$. The coordinate intersections now give every asserted formula. □

4 Exact source scope and attribution

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20, fixed twenty-page edition. R7 is discussed on p. 13 and again in Section 3.3.2; R9 is on p. 14. In R7, e is prime; R9 allows composite e. The proof above works in both cases. It supplies the conditional necessity missing from a proof of the canonical Constructions 5 and 7 alone. It proves neither the source's false assertion that these and the other listed residues exhaust all cases, nor a classification of unrestricted relative positions outside the prescribed residue.

There are also literal errors in the R7 paragraph. Its final affine group is called G_4 although the types are $0, 1, 2, 3$; it is G_3. Its printed condition $n' \mid (q-1)/(s'-1)$ does not express the required additional scalar factor. The correct condition is $n' \mid (s'-1)/(t-1)$, with $s' = v$, as printed in the R9 paragraph. The theorem gives the corrected full lattice without using either of these inconsistent expressions.

The classical ambient structure and conjugacy facts are borrowed as stated. The scalar valuation argument is reused from the accepted split-family proof. The coordinate containment lemma and the necessity argument above are the present contribution. They repair a gap in our first whole-source claim ledger, which confused a refuted global classification with these separate conditional assertions. No new finite search, Lean declaration, principal refutation, or compression measurement is claimed.